RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter 9B: Applications of Integration

1 **Do not use a calculator in answering this question**

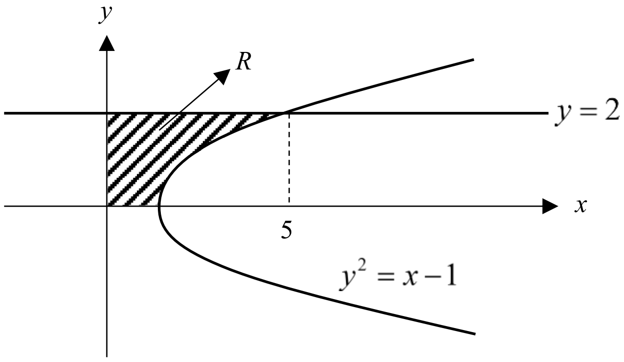
A curve has equation $y=\frac{3}{x}$ and a line has equation $y+2x=7$. The curve and the line intersect at the points A and B.

(i) Find the x-coordinates of A and B. [2]

(ii) Find the exact volume generated when the area bounded by the curve and the line is rotated about the x-axis through 360°. [4]

<table>
<tr><td colspan="2">9758/2018/01/Q2</td></tr>
<tr><td colspan="2">Solution:</td></tr>
<tr><td>(i)</td><td>When the curve intersects the line,
$$\frac{3}{x}=7-2x$$
$$2x^2-7x+3=0$$
$$(2x-1)(x-3)=0$$
$$x=\frac{1}{2} \text{ or } 3$$
Hence x-coordinates of the points A and B are $\frac{1}{2}$ and 3 respectively.</td></tr>
<tr><td>(ii)</td><td>Volume generated when region is rotated about the $x-$axis through 360°
$=\pi\int_{\frac{1}{2}}^{3}\left(y_1^{\,2}-y_2^{\,2}\right)\,\mathrm{d}x$
$=\pi\int_{\frac{1}{2}}^{3}\left((7-2x)^2-\left(\frac{3}{x}\right)^2\right)\,\mathrm{d}x$
$=\pi\left[\frac{(7-2x)^3}{-6}+\frac{9}{x}\right]_{\frac{1}{2}}^{3}$
$=\pi\left(-\frac{1}{6}+3+36-18\right)=\frac{125\pi}{6}\text{ units}^3$</td></tr>
</table>

2 The region R, in the first quadrant is bounded by the graph $y^2 = x-1$, the axes and the line $y=2$. Find the volume of the solid generated when R is rotated through one revolution about the **(a)** y-axis and **(b)** x-axis. [5]

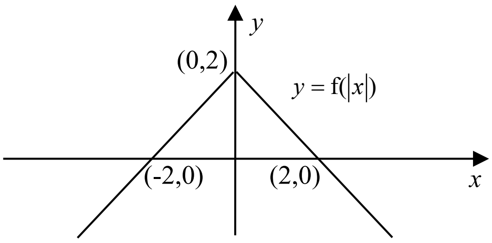
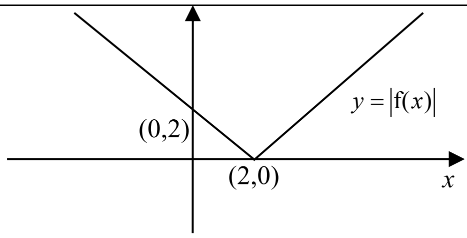
1985June/02/Q7(a) (Modified)	
Solution:	
(a)	 Required Volume $= \pi\int_0^2 x^2 \, \mathrm{d}y$ $= \pi\int_0^2 \left(y^2+1\right)^2 \mathrm{d}y = 43.1$ units^3 (3sf)
(b)	Required volume $= \pi\times 2^2\times 5 - \pi\int_1^5 y^2 \, \mathrm{d}x$ $= 20\pi - \pi\int_1^5 (x-1)\, \mathrm{d}x = 37.7$ units^3 (3sf) OR: Volume $= \pi\int_0^5 (y_1)^2 \, \mathrm{d}x - \pi\int_1^5 (y_2)^2 \, \mathrm{d}x$ $= \pi\int_0^5 (2)^2 \, \mathrm{d}x - \pi\int_1^5 x-1 \, \mathrm{d}x$

3 It is given that $\mathrm{f}(x) = 2 - x$.

(i) On separate diagrams, sketch the graphs of $y = \mathrm{f}\left(|x|\right)$ and $y = \left|\mathrm{f}\left(x\right)\right|$, giving the coordinates of any points where the graphs meet the x- and y-axes. You should label the graphs clearly. [3]

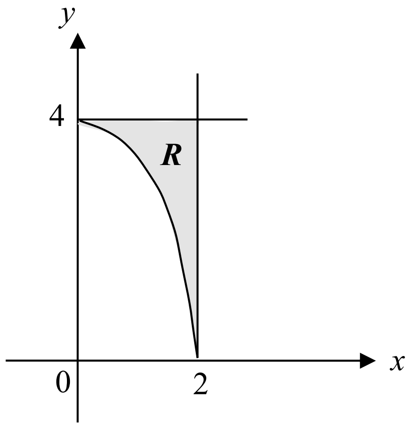
(ii) State the set of values of x for which $\mathrm{f}\left(|x|\right) = \left|\mathrm{f}\left(x\right)\right|$. [1]

(iii) Find the exact value of the constant a for which $\int_{-1}^{1} \mathrm{f}\left(|x|\right)\,\mathrm{d}x = \int_{1}^{a} \left|\mathrm{f}\left(x\right)\right|\,\mathrm{d}x$. [3]

9740/2011/01/Q5	
Solution:	
(i)	 
(ii)	$\{x \in \mathbb{R} : 0 \le x \le 2\}$ or [0, 2]
(iii)	$\int_{-1}^{1} \mathrm{f}(\lvert x\rvert)\ \mathrm{d}x = 2\int_{0}^{1} \mathrm{f}(\lvert x\rvert)\ \mathrm{d}x = 2\times\frac{1}{2}\times 1\times\left(2+1\right) = 3$. Since $\int_{1}^{2} \lvert \mathrm{f}(x)\rvert\ \mathrm{d}x = \frac{1}{2}\times 1\times 1 = \frac{1}{2} < 3$, so $a > 2$ $\int_{1}^{a} \lvert \mathrm{f}(x)\rvert\ \mathrm{d}x = \int_{-1}^{1} \mathrm{f}(\lvert x\rvert)\ \mathrm{d}x$ $\int_{1}^{2} \lvert \mathrm{f}(x)\rvert\ \mathrm{d}x + \int_{2}^{a} \lvert \mathrm{f}(x)\rvert\ \mathrm{d}x = 3$ $\frac{1}{2} + \frac{1}{2}\left(a-2\right)^2 = 3$ $a = 2 \pm \sqrt{5}$ Since $a > 2$, $a = 2 + \sqrt{5}$

4 The diagram shows the region R in the first quadrant bounded by the curve $x=\sqrt{4-y}$, and the lines $x=2$ and $y=4$.

Find , in terms of π, the volume of the solid obtained when R is rotated through 2π about the x-axis.



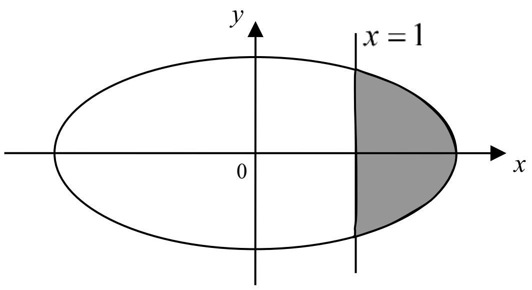
(i) Write down the equation obtained when the curve is translated 2 units in the negative x−direction. [1]

(ii) Hence find, in terms of π, the volume of the solid obtained when R is rotated through 2π about the line $x=2$. [4]

Solution:	
	Required volume of solid when R is rotated through 2π about the x-axis $=\pi\left(4^2\right)(2)-\pi\int_0^2 y^2\ \mathrm{d}x$ $=32\pi-\pi\int_0^2\left(4-x^2\right)^2\ \mathrm{d}x=\dfrac{224}{15}\pi$ OR: $\text{Volume}=\pi\int_0^2\left(y_1\right)^2\mathrm{d}x-\ \pi\int_0^2\left(y_2\right)^2\mathrm{d}x$ $=\pi\int_0^2(4)^2\ \mathrm{d}x\ -\ \pi\int_0^2\left(4-x^2\right)^2\ \mathrm{d}x$
(i)	Replace x in the original equation by $x+2$: $x+2=\sqrt{4-y}\quad\Rightarrow\quad x=\sqrt{4-y}-2$
(ii)	Required volume $=\pi\int_0^4 x^2\ \mathrm{d}y$ $=\pi\int_0^4\left(\sqrt{4-y}-2\right)^2\ \mathrm{d}y$ $=\pi\int_0^4\left(8-y-4\sqrt{4-y}\right)\ \mathrm{d}y$ $=\pi\left[8y-\dfrac{y^2}{2}+4\times\dfrac{(4-y)^{\frac{3}{2}}}{\frac{3}{2}}\right]_0^4$ $=\pi\left[32-\dfrac{4^2}{2}+\dfrac{8}{3}\left(0-4^{\frac{3}{2}}\right)\right]=\dfrac{8}{3}\pi\ \text{units}^3.$

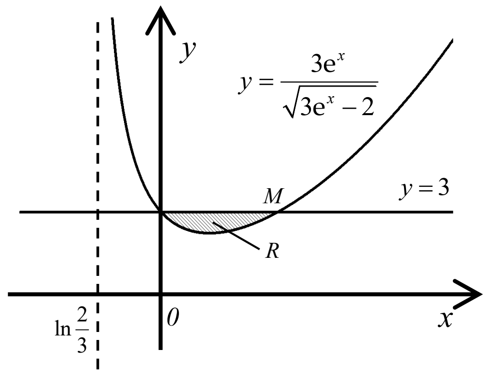
5 (a) The figure on the right shows the ellipse $\left(\frac{x}{2}\right)^2 + y^2 = 1$ and the line $x = 1$. By using the substitution $x = 2\sin\theta$, find the area of the shaded region, giving your answer in an exact form.

(b) Find the exact volume generated when the shaded region is rotated through 2π radians about the y-axis.



Solution:		
(a)	$\left(\frac{x}{2}\right)^2 + y^2 = 1 \Rightarrow y = \pm\sqrt{1-\left(\frac{x}{2}\right)^2}$ Required Area $= 2\int_1^2 \sqrt{1-\left(\frac{x}{2}\right)^2}\,\mathrm{d}x$ $= \int_1^2 \sqrt{4-x^2}\,\mathrm{d}x$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sqrt{4-4\sin^2\theta}\times 2\cos\theta\,\mathrm{d}\theta$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2\cos\theta \times 2\cos\theta\,\mathrm{d}\theta$ $= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} 2\left(1+\cos 2\theta\right)\,\mathrm{d}\theta$ $= \left[2\theta + \sin 2\theta\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$ $= \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$ units^2	When $y = 0,\ x = \pm 2$ When $x = 1,\ \theta = \frac{\pi}{6}$ When $x = 2,\ \theta = \frac{\pi}{2}$ $x = 2\sin\theta \quad \Rightarrow \quad \frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\cos\theta$
(b)	When $x = 1$, $y = \pm\frac{\sqrt{3}}{2}$ $\left(\frac{x}{2}\right)^2 + y^2 = 1 \Rightarrow x^2 = 4 - 4y^2$ Volume $= 2\left[\pi\int_0^{\frac{\sqrt{3}}{2}} x^2\,\mathrm{d}y - \pi(1)^2\left(\frac{\sqrt{3}}{2}\right)\right]$ $= 2\pi\int_0^{\frac{\sqrt{3}}{2}} \left(4-4y^2\right)\mathrm{d}y - \pi\sqrt{3}$ $= 8\pi\left[y - \frac{y^3}{3}\right]_0^{\frac{\sqrt{3}}{2}} - \pi\sqrt{3}$ $= 8\pi\left(\frac{\sqrt{3}}{2} - \frac{3\sqrt{3}}{8(3)}\right) - \pi\sqrt{3} = 2\pi\sqrt{3}$ units^3	OR: Volume $= 2\left[\pi\int_0^{\frac{\sqrt{3}}{2}} \left(x_1\right)^2\mathrm{d}y - \pi\int_0^{\frac{\sqrt{3}}{2}} \left(x_2\right)^2\mathrm{d}y\right]$ $= 2\left[\pi\int_0^{\frac{\sqrt{3}}{2}} \left(4-4y^2\right)\mathrm{d}y - \pi\int_0^{\frac{\sqrt{3}}{2}} (1)^2\mathrm{d}y\right]$

6 The diagram shows the region R bounded by the curve C with equation $y=\dfrac{3\mathrm{e}^x}{\sqrt{3\mathrm{e}^x-2}}$ and the line $y=3$. M is a point of intersection of C and the line $y=3$.



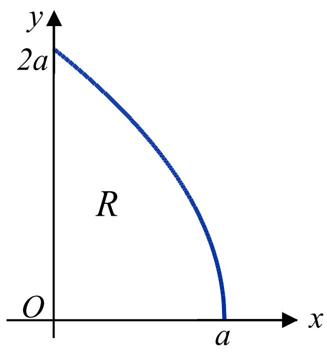
(i) Find the exact value of the x-coordinate of the point M. [2]

(ii) Show that the area of the region R is $3\ln 2-2$. [3]

(iii) Find the exact volume of the solid generated when R is rotated through 2π about the x-axis. (You may consider the use of the substitution $u=\mathrm{e}^x$) [6]

AJC Promo 9740/2006/Q12		
Solution:		
(i)	$\dfrac{3\mathrm{e}^x}{\sqrt{3\mathrm{e}^x-2}}=3$ $\mathrm{e}^x=\sqrt{3\mathrm{e}^x-2}$ $\mathrm{e}^{2x}=3\mathrm{e}^x-2 \;\Rightarrow\; \left(\mathrm{e}^x\right)^2-3\mathrm{e}^x+2=0 \;\Rightarrow\; \left(\mathrm{e}^x-1\right)\left(\mathrm{e}^x-2\right)=0 \;\Rightarrow\; x=0 \text{ or } \ln 2$ $\therefore$ $x-$coordinate of M is $\ln 2$.	
(ii)	Area of $R=\displaystyle\int_0^{\ln 2} 3-\frac{3\mathrm{e}^x}{\sqrt{3\mathrm{e}^x-2}}\,\mathrm{d}x$ $=\left[3x-2\sqrt{3\mathrm{e}^x-2}\right]_0^{\ln 2}=\left(3\ln 2-4\right)-\left(0-2\right)=3\ln 2-2$ (shown)	
(iii)	$\text{Volume}=\pi\left(3\right)^2\ln 2-\pi\displaystyle\int_0^{\ln 2}\frac{9\mathrm{e}^{2x}}{3\mathrm{e}^x-2}\,\mathrm{d}x$ $=9\pi\ln 2-\pi\displaystyle\int_1^2\frac{9u^2}{3u-2}\cdot\frac{1}{u}\,\mathrm{d}u$ $=9\pi\ln 2-\pi\displaystyle\int_1^2 3+\frac{6}{3u-2}\,\mathrm{d}u$ $=9\pi\ln 2-\pi\left[3u+\dfrac{6}{3}\ln\lvert 3u-2\rvert\right]_1^2$ $=9\pi\ln 2-\pi\left[3\left(2-1\right)+2\left(\ln 4-\ln 1\right)\right]$ $=9\pi\ln 2-\pi\left[3+2\ln 4\right]$ $=9\pi\ln 2-\pi\left(3+4\ln 2\right)=\pi\left(5\ln 2-3\right)$ units^3	$u=\mathrm{e}^x$ $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\mathrm{e}^x$ $\dfrac{\mathrm{d}u}{\mathrm{d}x}=u$ $\mathrm{d}x=\dfrac{1}{u}\,\mathrm{d}u$

7 The diagram shows the region R, which is bounded by the axes and the part of the curve $y^2 = 4a(a-x)$ lying in the first quadrant. Find, in terms of a,



(i) the area of R, [3]

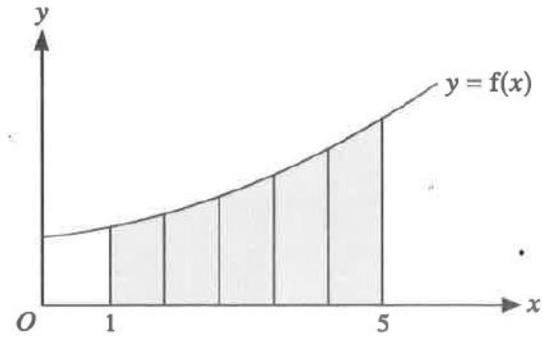
(ii) the volume V_x, of the solid formed when R is rotated completely about the x−axis. [2]

The volume of the solid formed when R is rotated completely about the y−axis is V_y. Show that $V_y = \frac{8}{15}V_x$.

The region S, lying in the first quadrant, is bounded by the curve $y^2 = 4a(a-x)$ and the lines $x = a$ and $y = 2a$. Find, in terms of a, the volume of the solid formed when S is rotated completely about the y−axis. [6]

9205/J92/01/Q16	
Solution:	
(i)	Area of R $= \int_0^a \sqrt{4a(a-x)}\ \mathrm{d}x = \sqrt{4a}\left[-\frac{2}{3}(a-x)^{\frac{3}{2}}\right]_0^a = \sqrt{4a}\left(\frac{2}{3}a^{\frac{3}{2}}\right) = \frac{4}{3}a^2$ units^2. OR: Area of R $= \int_0^{2a} x\ \mathrm{d}y = \int_0^{2a}\left(a - \frac{y^2}{4a}\right)\mathrm{d}y = \left[ay - \frac{y^3}{12a}\right]_0^{2a} = 2a^2 - \frac{2a^3}{3a} = \frac{4}{3}a^2$.
(ii)	$V_x = \pi\int_0^a 4a(a-x)\ \mathrm{d}x$ $= 4a\pi\left[-\frac{1}{2}(a-x)^2\right]_0^a$ $= 2\pi a^3$ units^3.
	$V_y = \pi\int_0^{2a} x^2\ \mathrm{d}y$ $= \pi\int_0^{2a}\left(a - \frac{y^2}{4a}\right)^2\ \mathrm{d}y$ $= \pi\int_0^{2a}\left(a^2 - \frac{y^2}{2} + \frac{y^4}{16a^2}\right)\mathrm{d}y$ $= \pi\left[a^2 y - \frac{y^3}{6} + \frac{y^5}{80a^2}\right]_0^{2a}$ $= \pi\left(2a^3 - \frac{4}{3}a^3 + \frac{2}{5}a^3\right) = \frac{16}{15}\pi a^3 = \frac{8}{15}V_x$ Required volume $= \pi a^2(2a) - V_y = 2\pi a^3 - \frac{16}{15}\pi a^3 = \frac{14}{15}\pi a^3$ units^3.

8 The diagram shows a sketch of the curve $y = \mathrm{f}(x)$. The region under the curve between $x = 1$ and $x = 5$, shown shaded in the diagram, is A. This region is split into 5 vertical strips of equal width, h.



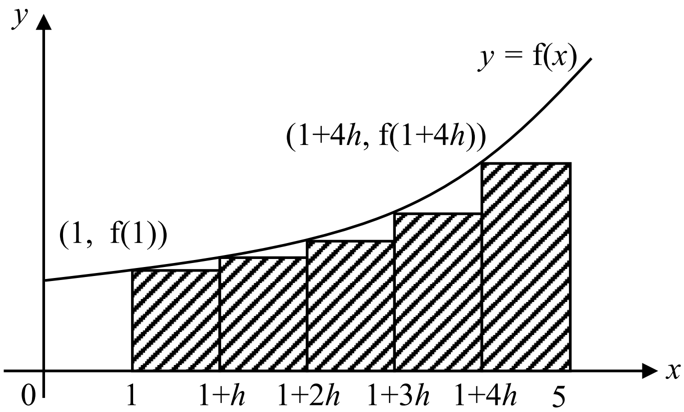
(a) State the value of h and show, using a sketch, that $\sum_{n=0}^{4}\left(\mathrm{f}(1+nh)\right)h$ is less than the area of A. [2]

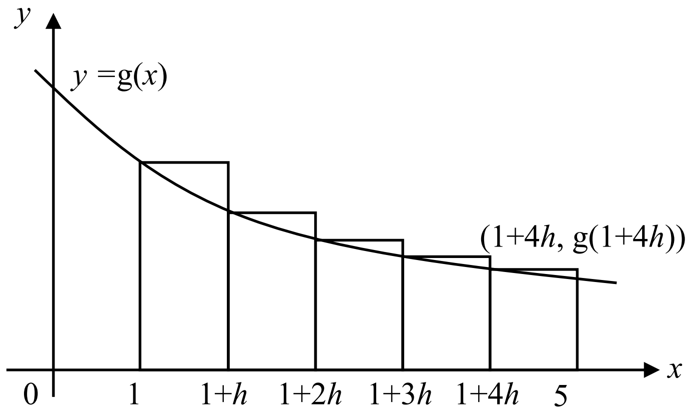
(b) Find a similar expression that is greater than the area of A. [1]

You are now given that $\mathrm{f}(x) = \frac{1}{20}x^2 + 1$.

(c) Use the expression given in part **(a)** and your expression from part **(b)** to find lower and upper bounds for the area of A. [2]

(d) Sketch the graph of a function $y = \mathrm{g}(x)$, between $x = 1$ and $x = 5$, for which the area between the curve, the x-axis and the lines $x = 1$ and $x = 5$ is less than $\sum_{n=0}^{4}\left(\mathrm{g}(1+nh)\right)h$. [1]

9758/2021/02/Q2	
Solution:	
(a)	$h = \frac{5-1}{5} = \frac{4}{5} = 0.8$  Total area of the 5 rectangles $=\mathrm{f}(1+0h)\times h+\mathrm{f}(1+h)\times h+\mathrm{f}(1+2h)\times h+\mathrm{f}(1+3h)\times h+\mathrm{f}(1+4h)\times h$ $=\sum_{n=0}^{4}\left(\mathrm{f}(1+nh)\right)h$ From the diagram, $\sum_{n=0}^{4}\left(\mathrm{f}(1+nh)\right)h$ is less than the area of A.

(b)	A similar expression that is greater than the area of A is $\sum_{n=1}^{5}\left(\mathrm{f}(1+nh)\right)h$
(c)	Lower bound for the area of A $=\sum_{n=0}^{4}\left[\left(\frac{1}{20}\left(1+n\left(\frac{4}{5}\right)\right)^2+1\right)\left(\frac{4}{5}\right)\right]=\frac{701}{125}=5.608$ Upper bound for the area of A $=\sum_{n=1}^{5}\left[\left(\frac{1}{20}\left(1+n\left(\frac{4}{5}\right)\right)^2+1\right)\left(\frac{4}{5}\right)\right]=\frac{821}{125}=6.568$
(d)	 Area between the curve, the x-axis and the lines $x = 1$ and $x = 5$ is less than $\sum_{n=0}^{4}\left(\mathrm{g}(1+nh)\right)h$ which is the total area of the 5 rectangles in the diagram.

9 **(a)** A flat novelty plate for serving food on is made in the shape of the region enclosed by the curve $y = x^2 - 6x + 5$ and the line $2y = x - 1$. Find the area of the plate. [4]

(b) A curved container has a flat circular top. The shape of the container is formed by rotating the part of the curve $x = \dfrac{\sqrt{y}}{a - y^2}$, where a is a constant greater than 1, between the points $(0,0)$ and $\left(\dfrac{1}{a-1}, 1\right)$ through 2π radians about the $y-$axis.

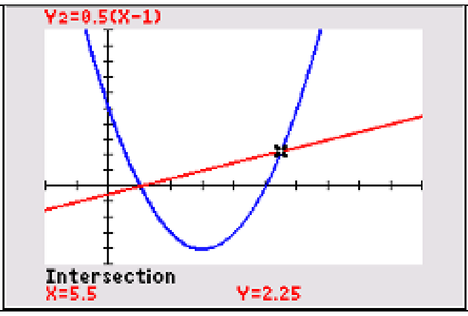
(i) Find the volume of the container, giving your answer as a single fraction in terms of a and π. [4]

(ii) Another curved container with a flat circular top is formed in the same way from the curve $x = \dfrac{\sqrt{y}}{b - y^2}$ and the points $(0,0)$ and $\left(\dfrac{1}{b-1}, 1\right)$. It has a volume that is four times as great as the container in part **(i)**. Find an expression for b in terms of a. [3]

9758/2017/02/Q4

Solution:

(a)

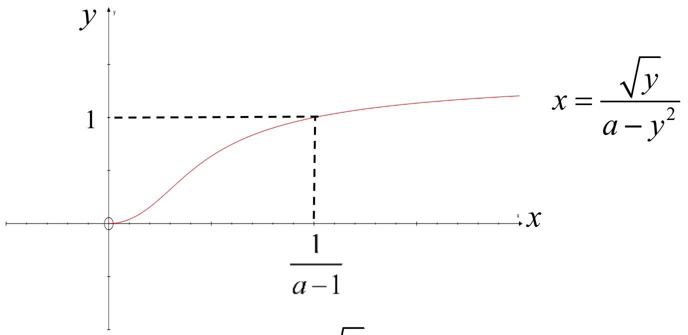


From GC, the x-coordinates of the points of intersections between the 2 curves are $x = 1$ and $x = 5.5$.

$$\text{Area of plate} = \int_1^{5.5} \frac{1}{2}(x-1) - \left(x^2 - 6x + 5\right) \mathrm{d}x$$

$$= 15.1875 \text{ units}^2 \text{ (exact) or } 15\frac{3}{16} \text{units}^2$$

(b) **(i)**



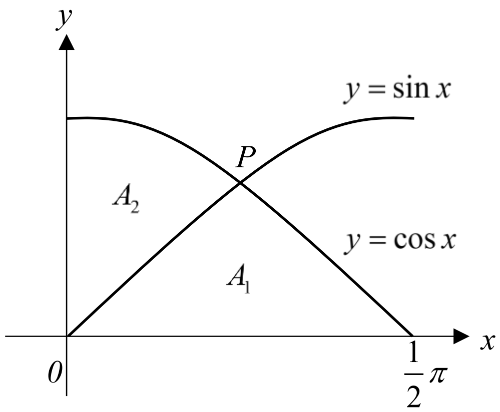
Note: The sketch of $x = \dfrac{\sqrt{y}}{a - y^2}$ can be obtained by sketching $y = \dfrac{\sqrt{x}}{a - x^2}$ and then reflecting it about the line y=x, since they are inverse functions of one another.

$$\text{Volume} = \pi \int_0^1 \left(\frac{\sqrt{y}}{a - y^2} \right)^2 \mathrm{d}y = \pi \int_0^1 \frac{y}{\left(a - y^2\right)^2} \mathrm{d}y = \frac{\pi}{-2} \int_0^1 \frac{-2y}{\left(a - y^2\right)^2} \mathrm{d}y$$

$$= \frac{\pi}{2}\left[\frac{1}{\left(a - y^2\right)} \right]_0^1 = \frac{\pi}{2}\left(\frac{1}{a-1} - \frac{1}{a} \right) = \frac{\pi}{2a(a-1)} \quad \text{units}^3$$

(b) (ii)	$\dfrac{\pi}{2b(b-1)}=\dfrac{4\pi}{2a(a-1)}$ $a^2-a=4b^2-4b$ $4b^2-4b+\left(a-a^2\right)=0$ $b=\dfrac{4\pm\sqrt{16-4(4)\left(a-a^2\right)}}{8}$ $=\dfrac{4\pm4\sqrt{1-a+a^2}}{8}$ $=\dfrac{1}{2}\left(1\pm\sqrt{1-a+a^2}\right)$ Since $a>1$, so $b>1$. Hence $b=\dfrac{1}{2}\left(1+\sqrt{a^2-a+1}\right)$

10 Do not use a calculator in answering this question.



With origin O, the curves with equations $y=\sin x$ and $y=\cos x$, where $0 \le x \le \frac{1}{2}\pi$, meet at the point P with coordinates $(\frac{1}{4}\pi, \frac{1}{2}\sqrt{2})$. The area of the region bounded by the curves and the x-axis is A_1 and the area of the region bounded by the curves and the y-axis is A_2 (see diagram).

(i) Show that $\dfrac{A_1}{A_2}=\sqrt{2}$. [4]

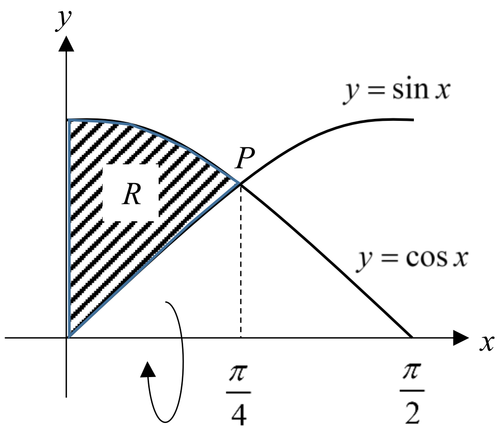
(ii) The region bounded by $y=\sin x$ between O and P, the line $y=\frac{1}{2}\sqrt{2}$ and the y-axis is rotated about the y-axis through $360°$. Show that the volume of the solid formed is given by

$$\pi\int_0^{\frac{1}{2}\sqrt{2}} (\sin^{-1} y)^2 \, dy.$$ [2]

(iii) Show that the substitution $y=\sin u$ transforms the integral in part **(ii)** to $\pi\int_a^b u^2 \cos u \, du$, for limits a and b to be determined. Hence find the exact volume. [6]

(iv) The region R is bounded by the y−axis and the curves $y=\sin x$ and $y=\cos x$ for $0 \le x \le \dfrac{\pi}{4}$. Find the volume of the solid of revolution formed when R is rotated completely about the x−axis, giving your answer in an exact form. [3]

9740/2015/01/Q10	
Solution:	
(i)	Since A_1 is symmetric about the line $x=\dfrac{1}{4}\pi$, $A_1 = 2\int_0^{\frac{\pi}{4}} \sin x \, dx$ $= 2\left[-\cos x\right]_0^{\frac{\pi}{4}}$ $= 2\left[-\dfrac{\sqrt{2}}{2}-(-1)\right] = 2-\sqrt{2}$

	$A_2 = \int_0^{\frac{\pi}{2}} \cos x \ \mathrm{d}x - A_1$ $= [\sin x]_0^{\frac{\pi}{2}} - \left(2-\sqrt{2}\right)$ $= 1-\left(2-\sqrt{2}\right) = \sqrt{2}-1$ $\dfrac{A_1}{A_2} = \dfrac{2-\sqrt{2}}{\sqrt{2}-1} = \dfrac{\sqrt{2}\left(\sqrt{2}-1\right)}{\sqrt{2}-1} = \sqrt{2}$
(ii)	$y = \sin x \quad \Rightarrow \quad x = \sin^{-1} y$ Required volume $= \pi \int_0^{\frac{\sqrt{2}}{2}} x^2 \ \mathrm{d}y = \pi \int_0^{\frac{\sqrt{2}}{2}} \left(\sin^{-1} y\right)^2 \mathrm{d}y$
(iii)	$y = \sin u \quad \Rightarrow \quad \dfrac{\mathrm{d}y}{\mathrm{d}u} = \cos u \quad \Rightarrow \quad \mathrm{d}y = \cos u \ \mathrm{d}u$ $y = 0 \quad \Rightarrow \quad u = 0$ $y = \dfrac{\sqrt{2}}{2} \quad \Rightarrow \quad u = \dfrac{\pi}{4}$ $\pi \int_0^{\frac{\sqrt{2}}{2}} \left(\sin^{-1} y\right)^2 \ \mathrm{d}y = \pi \int_0^{\frac{\pi}{4}} u^2 \cos u \ \mathrm{d}u$ (shown) where $a = 0$ and $b = \dfrac{\pi}{4}$ $= \pi \left\{ \left[u^2 \sin u\right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} 2u \sin u \ \mathrm{d}u \right\}$ $= \pi \left\{ \dfrac{\pi^2 \sqrt{2}}{32} - \left(\left[-2u \cos u\right]_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} 2\cos u \ \mathrm{d}u \right) \right\}$ $= \pi \left\{ \dfrac{\pi^2 \sqrt{2}}{32} + \dfrac{\pi\sqrt{2}}{4} - \left[2\sin u\right]_0^{\frac{\pi}{4}} \right\}$ $\therefore$ Volume $= \pi \left\{ \dfrac{\pi^2 \sqrt{2}}{32} + \dfrac{\pi\sqrt{2}}{4} - \sqrt{2} \right\}$ units^3
(iv)	Volume of solid formed when R is rotated completely about the x-axis $= \pi \int_0^{\frac{\pi}{4}} \left(\cos^2 x - \sin^2 x\right) \ \mathrm{d}x = \pi \int_0^{\frac{\pi}{4}} \cos 2x \ \mathrm{d}x$ $= \pi \left[\dfrac{\sin 2x}{2}\right]_0^{\frac{\pi}{4}} = \dfrac{\pi}{2}\left(\sin\dfrac{\pi}{2} - \sin 0\right)$ $= \dfrac{\pi}{2}$ units^3 

11 Curve C_1 has equation $y = \dfrac{1}{x}$ and curve C_2 has equation $y = \dfrac{1}{\sqrt{2-x^2}}$. Region Q is bounded by line $x = \dfrac{1}{\sqrt{2}}$ and both curves C_1 and C_2.

(a) Find the exact area of region Q. [3]

(b) The region bounded by the line $y = 5$ and the curves C_1 and C_2 in the first quadrant is rotated about the x–axis by 2π radians. Find the volume of the solid formed. [3]

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Solution:

(a)

When the curves meet, $\dfrac{1}{x} = \dfrac{1}{\sqrt{2-x^2}}$

$x^2 = 2 - x^2$

$x = 1$ (since $x > 0$)

$\therefore$ Area of Region Q

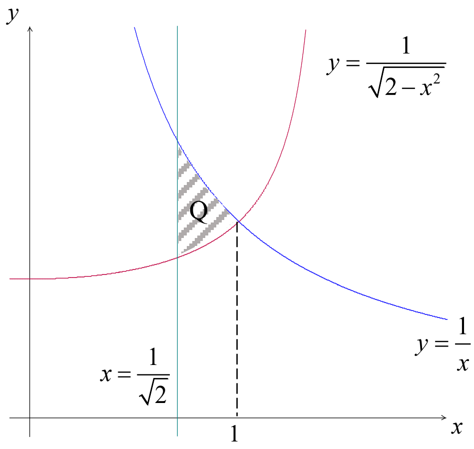
$$= \int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{x}\,\mathrm{d}x - \int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{\sqrt{2-x^2}}\,\mathrm{d}x$$

$$= \left[\ln x - \sin^{-1}\frac{x}{\sqrt{2}}\right]_{\frac{1}{\sqrt{2}}}^{1}$$

$$= \left(\ln 1 - \ln\frac{1}{\sqrt{2}}\right) - \left(\sin^{-1}\frac{1}{\sqrt{2}} - \sin^{-1}\frac{1}{2}\right)$$

$$= \ln\sqrt{2} - \frac{1}{4}\pi + \frac{1}{6}\pi$$

$$= \ln\sqrt{2} - \frac{1}{12}\pi \text{ units}^2 \quad \text{or} \quad \frac{1}{2}\ln 2 - \frac{\pi}{12} \quad \text{or} \quad -\frac{1}{2}\ln\frac{1}{2} - \frac{\pi}{12}$$



(b)

To find where the curves meet the line $y = 5$,

let $5 = \dfrac{1}{x}$ and $5 = \dfrac{1}{\sqrt{2-x^2}}$

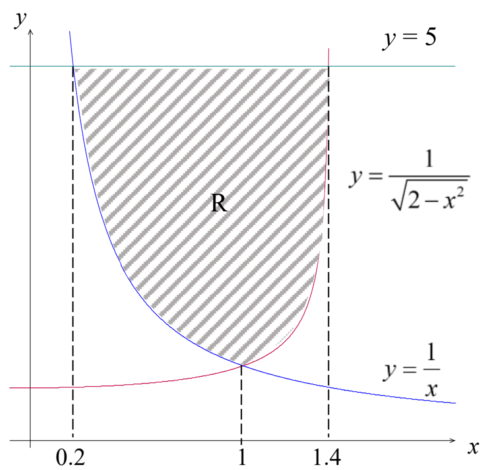
From GC, $x = 0.2$ and $x = 1.4$ (since $x > 0$)

Required volume

$$= \text{Vol of cylinder} - \pi\int_{0.2}^{1}\left(\frac{1}{x}\right)^2 \mathrm{d}x - \pi\int_{1}^{1.4}\left(\frac{1}{\sqrt{2-x^2}}\right)^2 \mathrm{d}x$$

$$= \pi(5)^2(1.4 - 0.2) - 12.566 - 3.9158$$

$$= 77.8 \text{ units}^3 \text{ (3sf)}$$



12 It is given that

$$\mathrm{f}(x)=\begin{cases}1, & \text{for } -1<x\le 0,\\ \dfrac{1}{1+4x^2}, & \text{for } 0<x\le 1,\end{cases}$$

and that $\mathrm{f}(x)=\mathrm{f}(x+2)$ for all real values of x.

(i) Find the exact value of $\mathrm{f}(23)$. [2]

(ii) Sketch the graph of $y=\mathrm{f}(x)$ for $-3\le x\le 3$. [3]

(iii) Find $\int_{-2}^{1}\mathrm{f}(x)\,\mathrm{d}x$, leaving your answer in exact form. [4]

(iv) The region bounded by the part of the curve $y=\mathrm{f}(x)$ where $0<x\le 1$, the line $y=\frac{1}{5}$ and the y-axis is rotated through 2π radians about the y-axis. Find the volume of the solid generated, giving your answer to three decimal places. [2]

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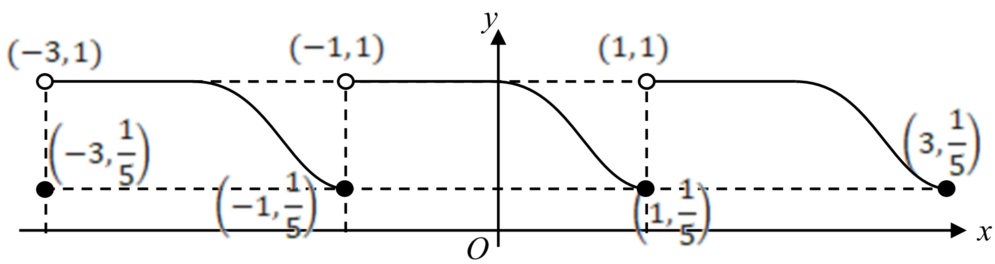
Solution:	
(i)	$\mathrm{f}(23)=\mathrm{f}(1)=\dfrac{1}{1+4}=\dfrac{1}{5}$
(ii)	
(iii)	$\int_{-2}^{1}\mathrm{f}(x)\,\mathrm{d}x=\int_{-2}^{-1}\mathrm{f}(x)\,\mathrm{d}x+\int_{-1}^{0}\mathrm{f}(x)\,\mathrm{d}x+\int_{0}^{1}\mathrm{f}(x)\,\mathrm{d}x$ $=\int_{0}^{1}\mathrm{f}(x)\,\mathrm{d}x + 1\times 1 + \int_{0}^{1}\mathrm{f}(x)\,\mathrm{d}x$ $=2\int_{0}^{1}\mathrm{f}(x)\,\mathrm{d}x + 1$ $=\int_{0}^{1}\dfrac{2}{1+(2x)^2}\,\mathrm{d}x + 1$ $=\left[\tan^{-1}(2x)\right]_0^1+1=\tan^{-1}(2)+1$
(iv)	$y=\dfrac{1}{1+4x^2} \Rightarrow x^2=\dfrac{1}{4y}-\dfrac{1}{4}$ Volume generated $=\pi\int_{\frac{1}{5}}^{1}\dfrac{1}{4y}-\dfrac{1}{4}\,\mathrm{d}y=0.636$ units^3 (3dp)